Advances in data networking

T J King

Over the past ten years or so, BT and a number of other telecommunications service providers have quickly become some of the largest players in the data networking arena. During this time networking technology has developed dramatically and shows no sign of slowing. This paper introduces the reader to data networking concepts, outlines the main areas of development in recent years and looks at some of the research and development areas currently under study in BT and around the world.

1. Introduction

The past decade has seen BT and a number of other telecommunications service providers quickly become dominant in the field of data networking. During this time networking technology has developed dramatically and this growth shows no sign of slowing. This paper introduces the main data networking concepts, describes the important areas of recent work within BT, and looks at some of the research and development currently under investigation in BT and world-wide.

1.1 Key concepts

Many terms that have now become commonplace in the data world can still appear daunting to someone relatively new to the subject. This section describes some of those terms and the concepts that are used widely in this field.

- Wide area network (WAN)

  This is a data network that interconnects a customer's geographically dispersed buildings or sites, usually by means of a public network service. Example WAN services are Switched Multi-megabit Data Service (SMDS), CellStream, Frame Relay and ISDN.

- Local area network (LAN)

  This is a data network of limited geographical size that typically operates at high speed and with very low bit error rate, normally contained within a building or a floor of a building, commonly used to connect desktop computers with each other and to share resources, e.g. printers and file servers. LANs are now frequently connected to other LANs, WANs or the Internet by means of devices known as routers and bridges.

- An internet

  Without a capital, this is short for internetwork, a collection of networks interconnected by routers that generally act as a single network — often confused with the Internet!

- Internet

  The Internet is the global internetwork that interconnects tens of thousands of networks and millions of hosts across the world. It evolved from the ARPANET which is discussed further in section 1.2. O'Neill et al [1] provide more details on the Internet and its development.

- Internetworking

  This is the science of interconnecting data networks (e.g. LANs) together by routers and other devices, so that the collection of networks thus created (normally) functions as a single network; products and technologies that allow such interconnection are known as internetworking products and technologies.

- Interworking

  This term refers specifically to the functions carried out at the boundary between two wide area networks (WANs), especially public networks, so that some form of interconnection is possible; it refers equally to voice or data networks. There are two flavours — network interworking and service interworking; for more detail on Frame Relay to ATM interworking see Walton [2].

- An intranet

An intranet is a private IP network typically employing the same technology for information retrieval as is used on the global Internet. Many organisations have their own corporate intranet with one or more connections to the Internet through an appropriate security ‘firewall’.

- Layers 1, 2, and 3

These layers are concepts defined by the International Standards Organisation (ISO) as part of the open systems interconnect (OSI) 7-Layer Reference Model [3]. Layer 1 refers to the physical layer, including the media specific items, e.g. optical fibre, transmitters and receivers, connectors and the line encoding schemes used. Layer 2, the data link layer, is broken into two sub-layers. The lower of these is known as the media access control (MAC) sub-layer and, perhaps not surprisingly, controls access to the physical layer, including access contention handling, and defines frame formats for use in the network (either LAN or WAN). The upper sub-layer defines the logical link control (LLC) which provides error control and flow control. The most widely used LLC is that defined in the IEEE 802.2 standard [4]. In its fullest implementation, LLC is quite complex; however, many practical instances are slimmed down or close to null. Layer 3, known as the network layer, is the first layer that handles end-to-end traffic and that has addressing with end-to-end significance. The Internet protocol (IP) and Novell’s Internet Packet Exchange (IPX) are essentially layer 3 protocols. Layer 3 describes the addressing, routeing and filtering functions required to ensure connectivity between end systems (computers). It also defines the format of the packets that make use of the frames provided by layer 2.

The OSI Reference Model has greatly helped to divide up the problem space and so allow different teams, through standards bodies and the like, to concentrate on specific parts of the so-called protocol stack. Similarly it has helped significantly as a structure around which to teach the basics of data networking — however, it is still only a model. The upper four layers of the model relate more to the end systems. The reader is referred to Day and Zimmermann [3] for more details.

- Router

Routers are multiple port network layer devices that handle layer 3 packets. They receive packets at one port, then act on the packet header information to route the packet via a particular output port to another router or end system. Routers may be interconnected across the wide area using a range of wide area services, e.g. Frame Relay, Switched Multi-megabit Data Service (SMDS), leased line, asynchronous transfer mode (ATM) and also the integrated services digital network (ISDN). They use routeing protocols or manually configured static routes to determine the routes.

- Routeing protocols

These are a set of data protocols employed by routers and some end systems that allow the paths across a network between end systems to be determined. They provide network reachability information through dynamic advertisement of routeing information. Routeing protocols are the rules which define the structure of routeing packets (the packets which routers use to exchange information about reachability of other networks) and how routers use these to build their internal routeing tables. These packets advertise routeing information and include some form of metric information to allow routers to calculate the optimum path to a particular destination network. The operation of dynamic routeing protocols is depicted in Fig 1.

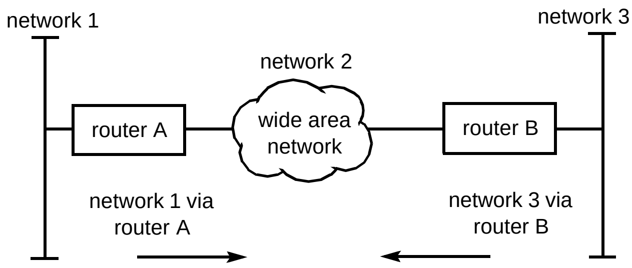

Fig 1 Dynamic routeing protocols in operation.

Routeing protocols can be split into two categories — distance vector and link state. Distance vector routeing protocols periodically advertise (send) their whole routeing table to a neighbouring router. Link state protocols on the other hand only forward updates when there is a change to the network topology.

Nevertheless, distance vector and link state protocols do provide the same reachability information. Examples of dynamic IP routeing protocols include the routeing information protocol (RIP) [5, 6], open shortest path first (OSPF) [7], and interior gateway routeing protocol (IGRP). More detail on routeing protocols and their deployment can be found in O’Neill et al [1], Whalley et al [8], Everrett et al [9] and Postel [10].

- Routed protocols

These are protocols that transport users’ data through an internetwork and can be routed by a router. They define the structures of packets which are routed through a network. The route taken by a packet depends upon the address, and routeing tables established in the routers. Examples of such protocols are the Internet protocol (IP) [10], Internetwork Packet Exchange (IPX) protocol (used by Novell NetWare), DECnet and AppleTalk.

1.2 A brief history of modern-day data networking

The explosion in data networking has been driven by a variety of related factors. In the late 1970s and early 1980s its early development in the US was led by research and experimentation on the packet-switched network known as the ARPANET. The need for survivable military communications networks funded some of the work and provided additional focus. The data protocol suite known as the transmission control protocol/Internet protocol (TCP/IP), that is now used world-wide, was originally developed and shared across the US as part of this effort.

At the same time a public data networking standard was being agreed at the international standards body, now known as the ITU-T, as Recommendation X.25. This was quickly adopted by many telecommunications operators as their first major public data network and still enjoys much success today, allowing geographically spread sites to be interconnected.

Devices known as modulators/demodulators, or modems for short, that allowed digital data to be transferred between pairs of sites across existing telephone networks and that operated within the voice frequency band, were also beginning to be used more widely.

In the 1970s the mainframe computer was in its heyday and mainframe vendors designed data protocols that allowed terminals to connect to the expensive (and hence scarce) mainframes. The most famous of these is IBM's systems network architecture (SNA), still in significant use today. Other manufacturers such as Digital Equipment Corporation, Xerox and Intel were solving other problems — how to connect their larger computers to their peripherals (such as tape and disk drives) and how to share expensive office automation (OA) resources such as printers and fileservers between several users. This work resulted in the definition of what is now the most prevalent of all local area networks (LANs) — Ethernet. In the early 1980s such LANs were initially stand-alone, though quickly the new challenge became how to interconnect two or more LANs. The LAN repeater and the LAN bridge — devices with increasing intelligence for LAN interconnection — were born.

At this stage in data communications, network management was something of the poor relation. By contrast, telecommunications networks already employed comparatively powerful network management.

Data network security was seen by data networking researchers as 'someone else's problem'. For example, in OSI Reference Model terms (described earlier), encryption of user data was seen as a presentation layer (layer 6) issue to be carried out in software on end systems, the network's function being simply to transfer whatever data it was given (encrypted or otherwise) reliably and accurately across the network. Encryption in software was relatively slow; more recently this has been vastly improved by hardware-assisted encryption devices operating at layers 1—3, providing much higher throughput.

By this time the basics of modern-day networking were established. Users now demanded larger internetworks that would work ever more reliably, particularly as levels of business-critical traffic carried over the networks increased. The introduction of more powerful software applications led to a need for higher bandwidths in the LAN. The need for more features and higher and more cost-effective bandwidth across the wide-area led to the development of a range of wide area network (WAN) services — Frame Relay, SMDS and ATM-based services such as BT's CellStream. These are described further in section 2.

The major drivers for advanced data networking are shown in Fig 2, with the shaded area representing the scope of this and the following papers.

2. Current issues in advanced data networking

2.1 The role of wide area network (WAN) services

BT offers a wide range of broadband and data services to its customers. Depending on the needs of the customer, these services vary in range from ATM, CellStream, through IP, Frame Relay, SMDS, ISDN and digital private circuits, to analogue leased lines which can be used for low bandwidth applications. Each of these services is provided on a separate platform which provides tailor-made solutions for that particular service; some examples follow.

- CellStream

  This is BT's newest broadband offering and is provided on an ATM platform. It allows the user to mix different traffic streams (voice, data, video) over a single transmission pipe. Access options are currently 2 Mbit/s, 34 Mbit/s and 155 Mbit/s. The experiences gained during the service pilot trial, combined with descriptions of the service and the network are described in Chauhan and Gallagher [11].

- BT Internet

  This is a value added service that provides customer access to the Internet via dial-in public switched telephone network (PSTN) or ISDN. BT also offers BTNet (the business Internet access service) aimed at high bandwidth users such as Internet service providers (ISPs).

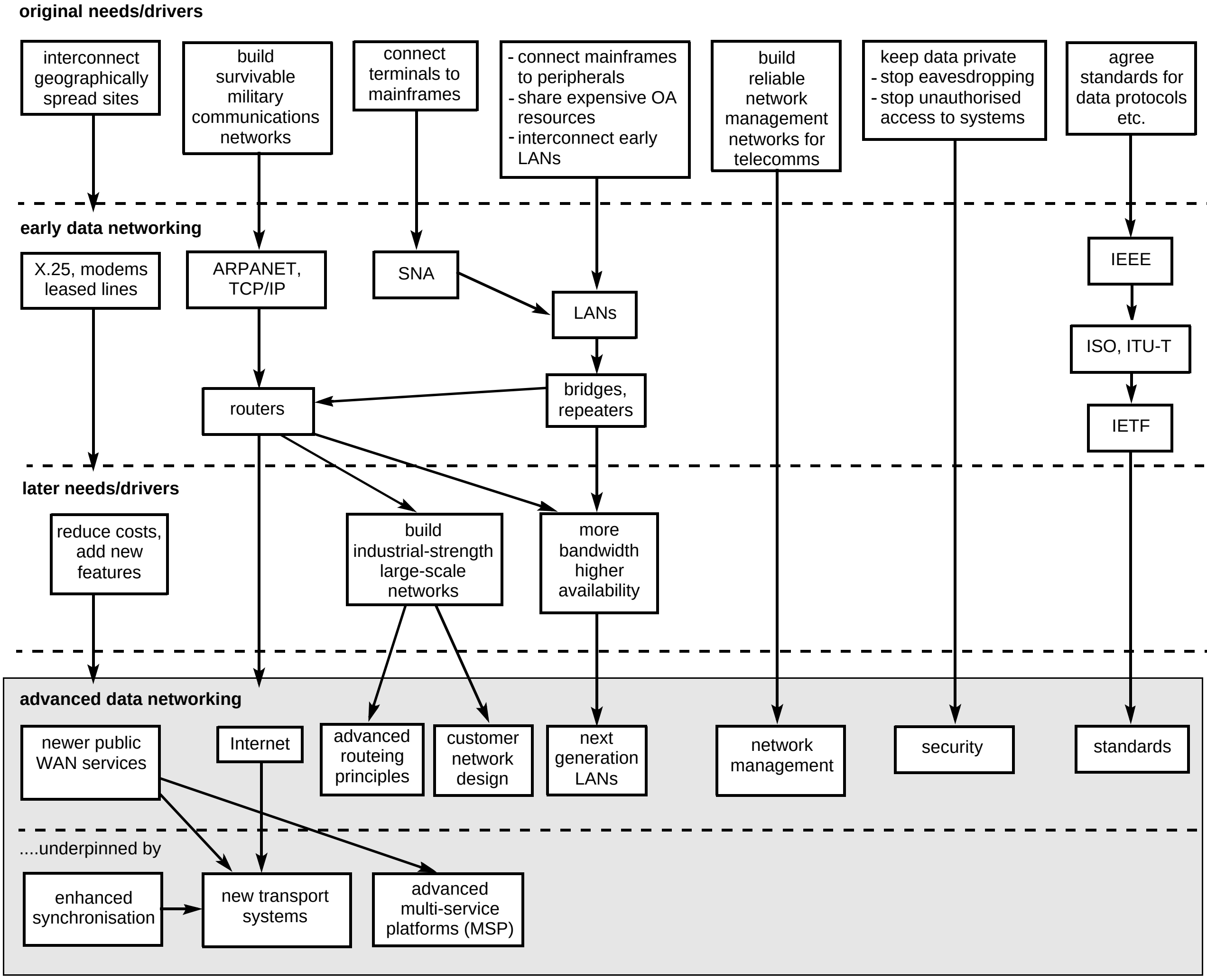

Fig 2 The drivers for advanced data networking (the shaded area represents the scope of this and the following papers).

- FrameStream (Frame Relay)

  This is a frame-based, connection-oriented service for customers needing bandwidth from 64 kbit/s up to 2 Mbit/s. As with most of these services, Frame Relay can be supplied from BT in the UK and from Concert as part of a global Frame Relay service. Some background to Frame Relay can be found in Walton [2].

- SMDS

  This Switched Multi-megabit Data Service is provided as a connectionless, frame-based data transport service. The service offers access classes from 0.2 Mbit/s up to 25 Mbit/s. SMDS is a LAN extension service, allowing users to interconnect their geographically separate LANs, and create a wide area LAN. The service was launched in December 1993 and is a major success, with over 2500 customer connections. The history and development of SMDS, along with a particularly interesting section on SMDS customers and their applications, can be found in Lewis et al [12].

- ISDN

  The integrated services digital network is a service offering access rates from 64 kbit/s up to 2 Mbit/s. Its role in data networking is described in detail in Everett et al [9]. ISDN is now used widely in major different ways in data networking:

  — to provide on-demand back-up paths for resilience in combination with another network service, for example with Frame Relay or SMDS,

  — to offer ‘top-up’ bandwidth, normally in conjunction with a leased line service such as KiloStream,

  — as a primary means of on-demand access for smaller sites.

- MegaStream

  The MegaStream service offers mainly 2 Mbit/s, but also 8, 34, 140 and 155 Mbit/s line rates, and is provided on a point-to-point basis. This point-to-point service lends itself well to customers who require high bandwidth between a small number of sites. A high performance MegaStream service, provided over the SDH and known as MegaStream Genus, is also available.

- KiloStream

  This service is the lower bandwidth version of MegaStream. Also point-to-point, it offers lines rates up to and including 64 kbit/s. KiloStream N supplies N × 64 kbit/s multiples, from 128 kbit/s up to 1024 kbit/s.

- Flexible bandwidth service (FBS)

  FBS is a voice and data service that offers tailored bandwidth to the end customer. Changes to the service can be arranged at very short notice.

As well as providing WANs for end customers, BT makes large use of them internally. One example is described in Cole et al [13], which outlines the data network used to manage BT's UK synchronous digital hierarchy (SDH) based transport network.

2.2 The multi-layered nature of networks and services

Figure 3 shows the multi-layered nature of networks and services in a new way. Any customer connection can be considered as a radial on the diagram. As an example, a customer buying BT's wide-area ATM service, CellStream, can be seen to be connected to BT's ATM network which in turn uses the SDH and PDH physical layer for transmission. This physical layer transmission is carried over the appropriate physical medium dependent (PMD) layer which, in turn, makes use of the physical infrastructure.

In the core of Fig 3 is the physical infrastructure used by all networks, including the physical ducting beneath the roads, etc, and the synchronisation that provides BT's network timing.

The next layer, or 'ring', out provides the PMD layer for the appropriate transmission types, e.g. optical fibre based for 34 Mbit/s plesiochronous digital hierarchy (PDH) and wavelength division multiplexing (WDM).

The 'SDH and PDH' layer provides an appropriate interface to the next layer out. It is this next layer where much (though not all) of the switching takes place. It includes the ISDN core network, the automatic crossconnect equipment (ACE), upon which BT's KiloStream service is based, and the metropolitan area network (MAN)/Cascade network that provides SMDS to customers.

The next layer out is predominantly the 'service' layer and includes CellStream, SMDS and so on. This is an important layer, as it is the outer edge of this layer that

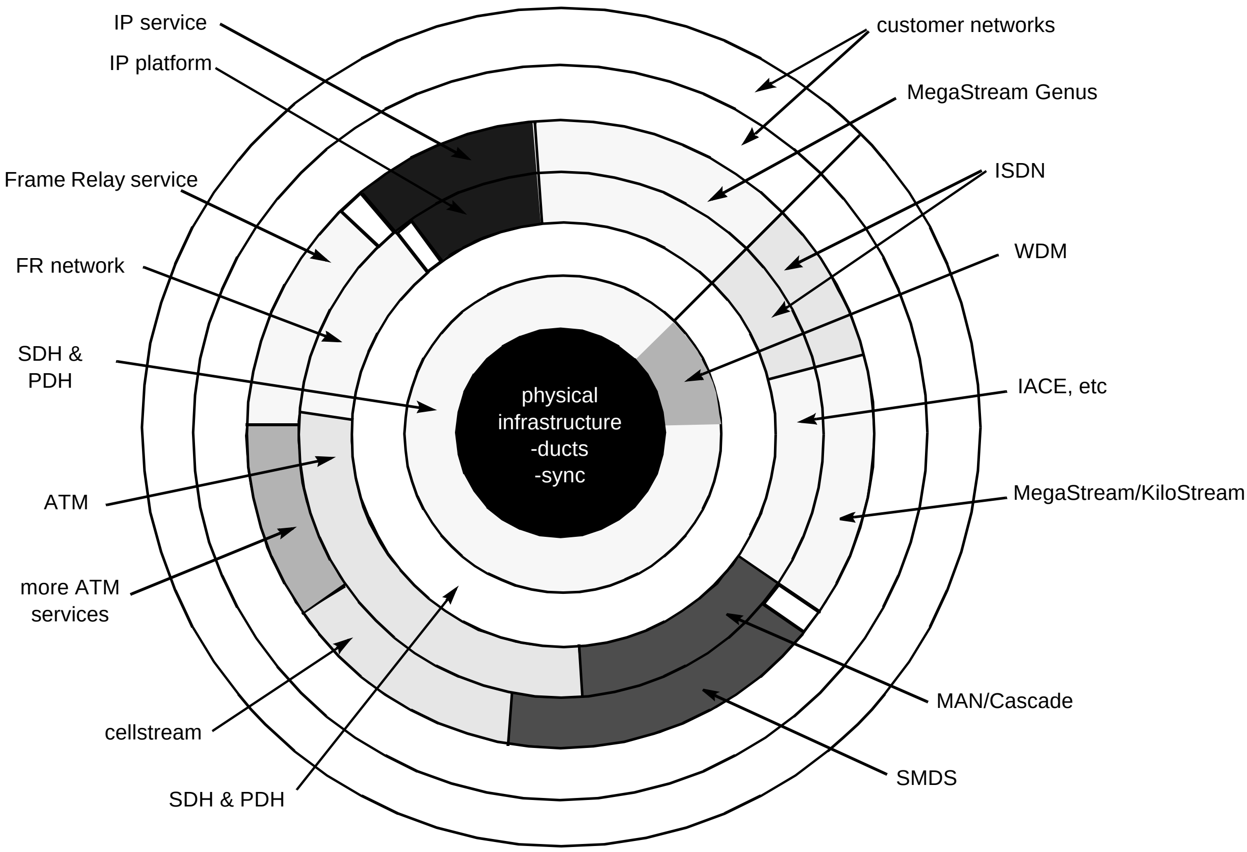

Fig 3 The multi-layered nature of networks and services.

defines the service provided to the customer. Section 2.1 described how, historically, each service has had its own platform. This is now blurring, as can be seen by the overlap between some of the services and the underlying platforms. So, for example, the ATM network is shown to be not only capable of providing CellStream and other ATM services, but also able to provide Frame Relay (FR) or SMDS over the same network. This leads to the concept of the multiservice platform, which potentially looks a very promising way to reduce the operational costs associated with running multiple, distinct platforms. The multiservice platform concept is discussed in Lewis et al [14].

A requirement of such a platform is the interworking between networks and services. In Walton [2], this is discussed for Frame Relay and ATM interworking.

The outer layers are part of the end customer's network. It should be noted that it surrounds all possible services and thus could make use of multiple services to create a customer data network or internetwork. A large bank, for example, may well use CellStream connected to its larger sites, Frame Relay to its branches, with ISDN to all sites as 'dial-around' back-up connectivity. The routers in the customer networks would choose the most appropriate routes through the internetwork. BT is very strong in the design of such internetworks, working closely with our customers. Some of the leading-edge design guidelines and the developments in this field are discussed in technical detail in Whalley et al [8].

2.3 Advanced data network design

Computer networks are growing ever larger, with more hosts and more sites. Networks with more than 1000 sites connected are not uncommon, encouraged by mergers, acquisitions and partnerships. At the same time more networked computers are being used at each site. In Whalley et al [8], techniques for achieving industrial-strength network designs are described. The paper includes a case study for a large-scale Windows NT design, then looks at new technologies that may help in the future.

One of the hottest topics in data network design at present is security. Security is very often a balance of the risks between making a service acceptable to users and making it secure from unauthorised users, for example, the time to commence a connection to a central facility — if you had to wait an hour while multiple layers of validation occurred before you could connect to your fileserver/e-mail post office, etc, then you would probably never use it. BT has significant experience in this area and Forrester et al [15] give an excellent tutorial on this wide-ranging subject.

2.4 Transport infrastructure for data networks

There has been something of a divide until recent years between data network designers and line transmission experts. Data experts have often started from building internetworks using a combination of routers and leased lines and traditionally have assumed only basic connectivity with relatively poor error rate performance from the WAN physical layers. The increased options for resilience at various network layers — for example, SDH fast protection switching — as well as the improved performance of SDH compared with earlier transmission systems, means that a much more effective solution can be designed by understanding the bigger picture and designing networks to suit. An overview of broadband transport, and particularly SDH is given in Brown et al [16]. An introduction to network synchronisation and its impact on data networking is outlined in Brown, Gilson and Mason [17].

3. Future developments in advanced data networking

3.1 Next generation LANs

New LANs are being developed to meet increasing user demands for more bandwidth, lower latency and increased reliability. Other requirements include the need to control OA support costs, support of two or more network layer protocols (e.g. IP and IPX), and the control of delay for use with time-sensitive applications.

Current LAN solutions include Ethernet 100baseT (with switched or shared media), the fibre distributed data interface (FDDI) and ATM, along with the current fast-moving developments on Gigabit Ethernet. Additional schemes are needed to make best use of some of these, e.g. user priority in bridged LANs and multi-protocol over ATM (MPOA) for ATM and virtual LANs.

These demands and the possible solutions to meet them are discussed in Newson et al [18].

3.2 Switched virtual circuits (SVCs) in ATM networks

Experiences with a pilot scheme of a new ATM service are described in Chauhan and Gallagher [11]. ATM has features that allow it to provide many exciting new services across the wide area. Many research and advanced development projects world-wide are evaluating ATM switched virtual circuits (SVCs). Unlike PVCs, which are set up at time of subscription, SVCs allow true dial-on-demand capability. Several papers in this special issue discuss ATM SVCs, each from a particular angle:

- an introduction to SVCs can be found in Clarke et al [19], which also describes the SVC signalling requirements and addressing requirements — the Joint

ATM Experiment on European Services, known as Project JAMES, is described with particular focus on the JAMES SVC trials across Europe,

- the application of ATM SVCs in LAN emulation (LANE) and multiprotocol over ATM is described in Newson et al [18],

- applying the IP subnet model to ATM SVC networks is discussed in Whalley et al [8], which outlines the challenges and discusses ways of overcoming them to allow large-scale ATM SVC-based internetworks to be designed — in particular it introduces the next hop resolution protocol (NHRP) as a means for providing shorter data paths (a technique known as ‘cut-through’) across the ATM network,

- a broadband call control demonstrator built at BT Laboratories that implements ATM signalling to the ITU-T Q.2931 Recommendation is described in Reece et al [20], who also cover the implementation of application and presentation layers.

4. Conclusions

Some of the history and key concepts of data networking have been described in this paper, which has also acted as an introduction to the following papers in this edition of the BT Technology Journal.

Developments over the past five years including the launch of new WAN services such as CellStream and SMDS, as well as enormous improvements in the features and reliability of routers, have moved data networking into a new age.

Acknowledgements

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Tim King has an MA in Engineering from Cambridge University and is in charge of Broadband and Data Networks, based at BT Laboratories, leading teams at both Martlesham Heath and in London. He joined BT Laboratories in 1988, initially working on ATM and then was technically responsible for the trial and pilot of BT’s switched multi-megabit data service (SMDS). He then led the internetworking team, working closely with end customers and account teams to provide world-class data internetworking solutions. Prior to BT he worked on high-speed data networking at BICC Data Networks (now part of 3COM) and on broadband telecommunications at the GEC Hirst Research Centre. He is a Chartered Engineer and a Member of the IEEE and IEE.